Can Decision Models Replace Tool Calling for Coding Agents?

Aziz Muminov Alexandra Ilyina

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Abstract

A coding agent connected to many MCP servers must either load every tool definition into its context or find tools through a search mechanism. Loading all 613 tools of twelve popular servers costs about 180k tokens per request, and current harnesses ship different search mechanisms: name lists with keyword search (Claude Code), BM25 (Codex, Anthropic's tool search, Hermes Agent), tool groups (GitHub Copilot), and embeddings or an LLM router (Goose). We ask whether a *decision model* (DM), which returns calibrated probabilities over up to 255 caller-supplied options without generating text, is a better tool selector than these. We built a test suite from the real tool definitions of 30 MCP servers (613 and 1,076 tools) with simulated back ends, 71 coding tasks graded by hidden tests and required writes, and two agent harnesses (opencode and Hermes Agent). Offline, on 120 queries, the DM ranks best (hit@1 0.93 at 613 tools against 0.48–0.54 for name, BM25 and embedding search and 0.86 for an LLM router, at a third of the router's cost); on queries written by agents the gap narrows. In live agents every search mechanism reaches ceiling pass rates (96–100% on 25 tool tasks) and cuts the largest prompt by 5–16× against loading all tools. Swapping BM25 for the DM inside Hermes's own search bridge used 12% fewer tokens, but the 95% interval of the ratio (0.77–1.02) includes no difference; how results are packaged (compact hits, typed calls) mattered more than how they were ranked. We report the methodology lessons that changed our conclusions: an unrestricted agent shell leaked the catalog and the ranker, provider routing added large variance, and only paired designs detect token differences of this size. The final controlled run (38 tasks, 1,076 tools, a pinned provider, 456 paired runs) is **[pending]**.

1 Introduction

Coding agents now reach far beyond the repository. Through the Model Context Protocol (MCP) [17] an agent can read CI logs, error trackers, tickets, chat, dashboards, clusters and databases. Each MCP server contributes tools, and each tool a definition (name, description, parameter schema) that the agent's LLM must see before it can call the tool. Twelve widely used servers (GitHub, Grafana, Sentry, Atlassian, Linear, Slack, Notion, Kubernetes, MongoDB, Postgres, Git, Playwright) expose 613 tools whose definitions take about 180k tokens. Loaded on every request, they fill most of a context window before the agent has read a line of code.

Harnesses have responded with *tool search*: the definitions stay out of the prompt and the agent finds tools on demand. The mechanisms differ (§2). Claude Code lists the names of deferred tools and loads them by exact name or keyword [3]; Anthropic's API and OpenAI's Codex CLI offer BM25 search [2, 20]; GitHub Copilot groups tools into "virtual tools" that the agent activates [14]; Goose ranks tools by embeddings or asks an LLM router [6]; Hermes Agent defers MCP tools behind a BM25 bridge [18]. Each is a ranking function over the catalog, and they have not been compared with each other on the same tools and tasks.

Decision models (DMs) are a natural candidate for this ranking. A DM reads a state and a question with named options and returns a probability for each option in one forward pass, without generating tokens; we use Microsoft-Decision-1 (MSD1) [15] through OpenRouter's Decisions API [22]. Choosing a tool from described options is exactly such a question. A DM is, however, limited to 255 options per question, fewer than a realistic catalog holds, and it cannot write the tool's arguments.

We therefore ask: *can a DM serve as the tool selector of a coding agent, and is it better than the mechanisms that harnesses ship?* To answer it we built a test suite on real tool definitions (§4): the actual `tools/list` output of 30 MCP servers (613 tools, and 1,076 with natural decoys such as GitLab beside GitHub), simulated back ends that answer from per-task world rules, 71 tasks in two versions graded by hidden tests, required writes and the absence of destructive calls, eight selection conditions, and two harnesses. Our findings, with the caveats of §6:

- **Offline, the DM is the most accurate ranker.** On 120 hand-written queries a server-then-tool DM hierarchy put the right tool first for 93% of queries at 613 tools and 88% at 1,076, against 45–54% for name, BM25 and embedding search and 82–86% for an LLM router, at about a third of the router's cost and less than half its latency (§5.1). On queries that agents actually wrote, the gap is smaller.
- **In live agents, every search mechanism is at ceiling.** With opencode, all catalog conditions passed 96–100% of runs on 25 tool tasks with two LLMs, and searching cut the largest prompt from about 182k tokens to 12–53k and the cost per run by up to 9× (§5.2). Pass rate cannot separate the mechanisms on these tasks; tokens and cost can, but only with many runs.
- **Inside one bridge, the DM's advantage is small and not yet significant.** In Hermes, replacing BM25 with the DM in the same bridge used 0.88× the tokens (95% CI 0.77–1.02) on 24 paired slots; a random ranker used about twice as many. How hits are packaged mattered more than how they are ranked (§5.3).
- **Measuring this is harder than it looks.** Our first Hermes comparisons were confounded by result packaging and by agents that used the shell to probe the machine, and repeated runs of the same setup differed by a median factor of 1.45 in tokens. We describe the fixes and the paired design that the final run uses (§5.3, §5.4).

Contributions. (i) A test suite for tool selection in coding agents built on real MCP tool definitions, with simulated back ends, graded tasks and a sandbox. (ii) A comparison of the mechanisms that current harnesses ship and of a DM, offline and inside two agents. (iii) Ways to use a DM beyond its 255-option limit. (iv) Methodology lessons for measuring token differences between agent configurations.

2 How harnesses select tools

Table 1 summarizes the mechanisms we compare. All except native loading share one pattern: the agent sees a small, fixed set of tools (a search tool and a way to call or load what it finds) plus, in most designs, a short overview of the servers, and pays for definitions only when it retrieves them.

Hermes's bridge. Hermes Agent 0.19.0 defers MCP tools once their definitions exceed 10% of the context window, which 613 tools do. Its bridge has three tools: `tool_search` returns compact

Table 1: Tool-selection mechanisms in current harnesses and the conditions that reproduce them.

Condition	Shipped in	What the agent sees and does
native	Cursor, Cline, opencode; Claude Code with search off	every tool definition in every request
cc	Claude Code ToolSearch [3]	names of all deferred tools; `select:a,b` by exact name, or keyword search with name matches weighted highest (our approximation; the exact scoring is undocumented)
bm25	Anthropic tool search [2], Codex CLI [20], Hermes [18]	Okapi BM25 [24] over names, descriptions and parameters
groups	GitHub Copilot virtual tools [14]	one `activate_<group>` tool per server (servers over 40 tools split); activating loads the group's definitions
embed	Goose vector router [6], RAG-MCP [10]	cosine similarity of `text-embedding-3-small` vectors of query and tool
llm	Goose LLM router [6]	an LLM (DeepSeek V4.1 Flash) reads one line per tool and names the best ones
dm	this paper	the DM ranks servers, then tools (§3)
random	control	five random tools

hits (name and one line), `tool_describe` returns one full definition, and found tools are then called directly with typed arguments. This packaging differs from the other search designs we implemented, which return five full definitions and route calls through a generic `call_tool(name, arguments)`. We return to this difference in §5.3.

Skills and progressive disclosure. Agent skills [1] apply the same idea to instructions: a one-line summary is always in context and the body is loaded on demand. The service overview that most search conditions include, and the "where things live" handbook of our v2 tasks (§4), play the role of such a summary for the catalog.

3 Decision models as tool selectors

The interface. A Decisions API request carries a *state* (here the agent's search query) and named questions; a `choice` question lists options, each with a free-text description, and returns a probability per option [22]. We describe each tool by its name, the start of its description (300 characters) and its parameter names. MSD1 is a post-trained 9B model [15]; it answers in about half a second and costs about $0.04 per million input tokens.

The 255-option limit. A `choice` question accepts at most 255 options, and MSD1 rejects requests beyond about 32k tokens. We compared four ways around the limit:

- **Hierarchy** (default): ask which server has the needed tool (12 or 30 options, each described by the server's purpose), take the most likely servers until they cover 90% of the probability mass (at most three), ask which of their tools fits (Linear, the largest server, has 198), and score each tool by $P(\text{server}) \cdot P(\text{tool} \mid \text{server})$.
- **Tournament**: split the catalog into chunks of at most 250, keep the top five of each, and decide among the survivors.

Table 2: The real tool catalog: tools per MCP server, from each server's own `tools/list`.

Catalog	Servers (tools)	Tools
main	Linear (198), Atlassian (98), GitHub (91), Grafana (81), Git (28), MongoDB (27), Playwright (25), Notion (24), Kubernetes (23), Sentry (9), Slack (8), Postgres (1)	613
extra	GitLab (127), monday (84), Salesforce (74), Heroku (33), HubSpot (21), AWS CloudWatch (19), AWS EKS (16), Airtable (16), Filesystem (14), CircleCI (13), Browserbase (9), Memory (9), Netlify (9), Terraform (9), Elasticsearch (4), ClickHouse (3), Figma (2), Docker (1)	+463
all	30 servers	1,076

- **BM25→DM** and **embed→DM**: shortlist 100 tools with BM25 or embeddings, then let the DM choose among them.

A variant redesigned for agent loops (*DM v2*) returns exact tool names found in the query without a DM call and truncates the hits once they cover 90% of the probability mass, so a confident DM returns one or two hits instead of five.

Why the DM selects and does not call. A DM chooses among options; it cannot produce the arguments of a call, and in a coding session most tool choices are easy. We checked this on 395 tool decisions of one real coding-agent session (the session in which this project was built). Given the user's request and the last eight steps, MSD1 predicted the agent's next tool among 38 with accuracy 0.49 (TypeSafe Jev [26]: 0.60), below the baseline "always the shell" at 0.75, although the DMs' top five contained the agent's tool 94–96% of the time. In the same session 51 always-loaded tool definitions took about 59k tokens, 93% of the first request and 16% of all context tokens the session read. The DM's place is therefore the catalog: deciding which of hundreds of rarely used tools to show, while the LLM keeps the core tools and writes every call.

4 Test suite

Real tool catalog. We started twelve MCP servers as an unprivileged user with dummy credentials and saved their `tools/list` responses: names, descriptions and input schemas exactly as the servers publish them (Table 2). No tool was ever executed against a real service. The resulting 613-tool catalog is our main setting. Eighteen further servers add 463 tools (1,076 in all). They include natural decoys for the main servers: GitLab beside GitHub, AWS CloudWatch beside Grafana Loki, AWS EKS beside Kubernetes.

Simulated back ends. The definitions are real; the answers are simulated. Each task has a world: a list of rules (tool, argument values to match, response), followed by organization-wide facts shared by all tasks (data sources, channels, teams). The first matching rule answers. Unmatched reads return what a real service returns for something that does not exist (an empty list, or not found); unmatched writes succeed and are recorded for grading. Optionally the simulator validates arguments against the tool's schema, as a real server would.

Tasks. Each task is a small Python repository whose fix needs information that lives only in the services: a failing CI check, a Sentry stack trace, Loki logs, a Prometheus alert threshold, a crash-looping pod, a Slack decision, a Jira or Linear ticket, a Confluence or Notion page, a MongoDB document shape, a Postgres schema. The sandbox has no network and no credentials, so the services

are reachable only through the catalog. Many tasks also require a write: comment on the pull request, resolve the Sentry issue, move the ticket, reply in the thread, open a pull request.

- **v1** (33 tasks): 22 single-service, 3 multi-service, 4 *absent* (the information does not exist; the right outcome is to stop and leave a `TODO`), and 4 controls that need no catalog tool. Prompts often name the service ("check the logs in Loki").
- **v2** (38 tasks, Table 7): the same worlds with *symptom-first* prompts that do not name the service ("customers stopped getting shipping e-mails"), a shared "where things live" handbook in every README (like an `AGENTS.md`, naming the company's tools, including legacy GitLab, CloudWatch and EKS), and five new tasks, four of them multi-service and three on the extra servers, so that the decoy servers are sometimes the right answer.

Every prompt ends with the same escape clause: if the information is not available, do not guess; leave the code unchanged and add a `TODO: info not found` comment. In the final run (§5.4) every prompt also starts, identically in all conditions, with a note that stands in for what real harnesses say in the system prompt about connected MCP servers: "Note: the company's internal services (code hosting, CI, error tracking, metrics and logs, incidents and on-call, tickets, docs and runbooks, chat, clusters, databases) are connected to you as tools. Some tools are loaded on demand: if a tool you need is not listed, use tool search to find it."

Grading and validation. A run passes when (1) the hidden tests pass, (2) every required write was made with the right arguments, and (3) no destructive call was made (merge, delete, rollout restart, `kubectl apply`, ...). Before any agent saw a task we checked that its hidden tests fail on the original repository and pass with the reference solution, that every tool it names exists in the catalog, and that the reference calls get non-empty answers from the simulator (empty ones for absent tasks) and satisfy the required writes.

Harnesses and sandbox. *opencode* [21] (v1.19.4) runs headless as an unprivileged user with an empty environment; its shell is limited to `pytest` and its file tools to the project, and the LLM is reached through a local proxy that holds the API key. The catalog is served by our MCP server, which implements every condition of Table 1; all ranked-search conditions return the top five tools with full definitions and a service overview, and calls go through `call_tool`. *Hermes Agent* [18] 0.19.0 has an unrestricted shell. In its default configuration it applies its own BM25 bridge to our MCP server's full tool list. To compare rankers within the same packaging we patched the bridge's search function to ask our server's `/rank` endpoint, which runs the DM (or a random ranker), when an environment variable is set. From the run `hermes_v3` on, each Hermes run has its own mount namespace (its own work directory, a fresh home directory and a private `/tmp`), and the MCP server answers only requests that carry a per-run token, which Hermes removes from its shell's environment (App. A).

5 Experiments

5.1 Offline retrieval

We wrote two queries for each needed step of the v1 tasks, one in natural language and one as keywords (120 in all), and counted a hit when the top k contained any acceptable tool for that step. Table 3 and Figure 1 give the results.

Table 3: Offline retrieval on 120 hand-written queries. Hit@k: a right tool is among the top k. Cost per 1,000 queries and median latency of the selecting call at 613 tools; name, BM25 and embedding search run locally (embedding cost is negligible).

	613 tools			1,076 tools			
Mechanism	hit@1	hit@5	MRR	hit@1	hit@5	$/1k queries	latency (s)
Claude Code-style (cc)	0.48	0.71	0.56	0.45	0.65	0	0.01
BM25	0.54	0.75	0.62	0.49	0.76	0	<0.01
embeddings	0.51	0.80	0.61	0.47	0.75	≈0	0.67
LLM router	0.86	**0.98**	0.92	0.82	**0.98**	0.44	1.31
DM hierarchy	**0.93**	0.97	**0.95**	**0.88**	0.97	0.15	0.54
DM tournament	0.90	**0.98**	0.94	0.86	**0.98**	1.13	1.60
BM25→DM	0.87	**0.98**	0.91	0.80	0.96	0.18	0.33
embed→DM	0.88	**0.98**	0.93	0.80	**0.98**	0.18	0.99

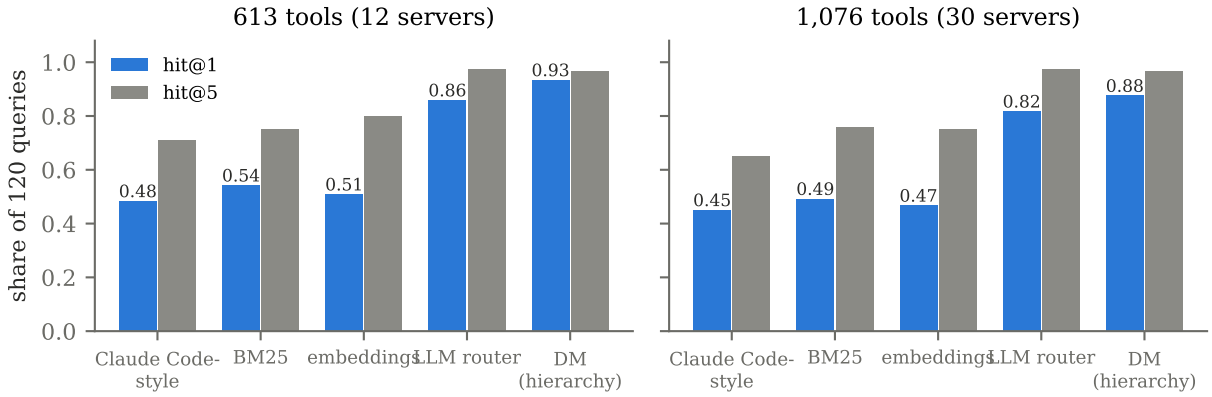


Figure 1: Offline retrieval, hit@1 (labelled) and hit@5, on the main and the extended catalog.

The DM ranks best; the lexical and embedding searches that harnesses ship rank worst. At 613 tools the DM hierarchy put a right tool first for 93% of queries; the mechanisms of Claude Code, Codex and Goose's vector router managed 48–54%, and the LLM router 86%. All learned rankers reached hit@5 of 0.97–0.98, against 0.71–0.80 for the others. The DM hierarchy cost $0.15 per thousand queries, a third of the LLM router, at less than half its latency; the tournament was slightly less accurate at hit@1 and seven times as expensive. Adding 463 tools with decoy servers lowered hit@1 by 3–6 points for every mechanism and left the order unchanged (DM 0.875, LLM router 0.82, BM25 0.49).

On queries written by agents the gap narrows. Agents do not write tidy queries. On 183 queries taken from our live runs, mostly lists of keywords and tool-name fragments, the DM still led at hit@1 (0.60 against 0.45 for BM25) but barely at hit@5 (0.70 against 0.66). Keyword-style queries suit lexical search, and agents compensate for a weak first query by searching again. This is the first hint that offline ranking quality does not translate directly into agent behaviour.

Table 4: opencode on task set v1, 613 tools (two repeats per task). Pass: runs passed on the 25 tasks that need a catalog tool. Largest prompt: median over those runs. Cost: mean LLM cost per run over all 33 tasks, excluding the selector (LLM router: $0.0011/$0.0031 per run; DM: $0.0003/$0.0010).

	DeepSeek V4.1 Flash			Tencent hy4		
Condition	Pass	Largest prompt	$/run	Pass	Largest prompt	$/run
none	0/25	9.6k	0.0110	0/25	15.4k	0.0154
native	25/25	181.9k	0.0722	25/25	188.5k	0.1039
cc	25/25	16.2k	0.0089	25/25	32.0k	0.0375
bm25	25/25	13.4k	0.0093	25/25	33.6k	0.0417
groups	25/25	29.0k	0.0179	24/25	52.7k	0.0562
embed	25/25	13.3k	0.0090	23/25	49.0k	0.0514
llm	25/25	11.7k	0.0076	25/25	32.6k	0.0419
dm	25/25	14.0k	0.0082	25/25	34.6k	0.0470

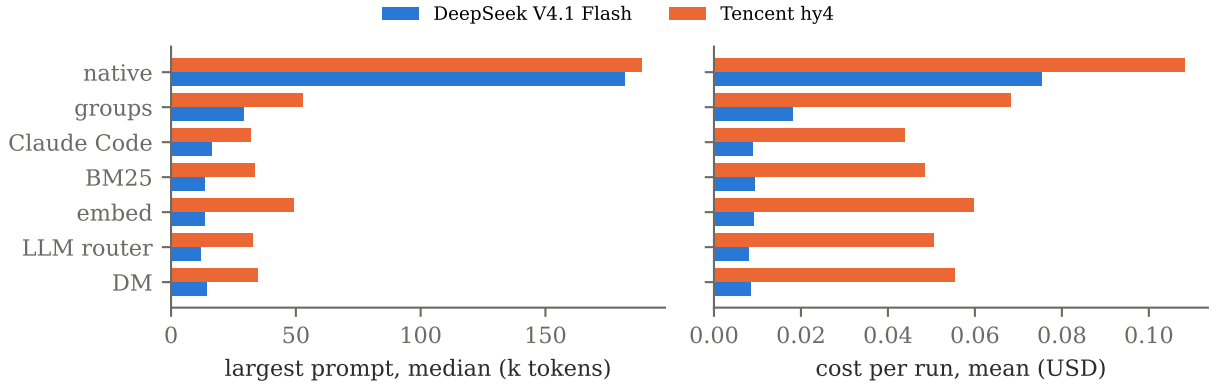


Figure 2: opencode, task set v1, 613 tools: median largest prompt on the 25 tool tasks and mean cost per run, by condition and LLM.

5.2 Live runs with opencode

Each of the 33 v1 tasks ran under eight conditions with two LLMs, DeepSeek V4.1 Flash [8] and Tencent hy4, twice for each pair of task and model (528 runs, 613-tool catalog; results `real_v1`). Table 4 and Figure 2 summarize them.

The tasks need the catalog, and every mechanism provides it. Without a catalog no tool task passed (0 of 50). With one, every condition passed 96–100%: native, cc, bm25, llm and dm 50 of 50, groups 49 and embed 48. All conditions abstained correctly on the four absent tasks (8 of 8 runs each), and no control run made a single catalog call. All three failures were hy4 runs: twice (groups, embed) the agent fixed the code but also restarted a production Kubernetes deployment it had not been asked to touch, which our grading counts as destructive, and once (embed) it stopped searching before it had found the ticket's specification.

Search removes most of the context. Loading every definition made the largest prompt about 182–189k tokens; with ranked search it was 12–16k (DeepSeek) and 32–35k (hy4), and 29–53k with groups, whose activation loads a whole server. That is 5–16× less, and the cost per run fell from $0.072 to $0.008–0.009 for DeepSeek and from $0.104 to $0.038–0.047 for hy4. Among the ranked

searches the differences in tokens and cost are within run-to-run noise (§5.3).

Runs use few tools. Across catalog runs the median run used 4 distinct catalog tools in about 10 calls (Hermes runs: 5 tools in 9–12 calls), almost always from one server; 23 of the 25 v1 tool tasks need only one server. Selecting the right server is therefore most of the problem, which suits the DM hierarchy but also makes the task easy for any mechanism that shows a service overview. Task set v2 was built to make it harder (§4).

5.3 Live runs with Hermes Agent

Hermes ships a different packaging from our MCP-side search, so it lets us ask how much of a mechanism's cost comes from the ranker and how much from the way hits reach the agent.

Two confounds in the first comparisons. Our first Hermes runs compared Hermes's built-in bridge with our MCP-side search conditions and could not be interpreted, for two reasons that we report as lessons:

1. *Packaging, not ranking.* Our MCP search returned five full definitions per query and routed calls through a generic `call_tool`. Its search results were about four times as large as Hermes's compact hits, and agents made 4.6 failed calls per run against 0.9 with typed direct calls. Any ranker plugged into it looked worse than BM25 in Hermes's bridge.
2. *Leakage through the shell.* Hermes's shell is unrestricted. In 40–60% of runs per condition the agent probed its environment: it sent requests to the MCP server's port, dumped the catalog, read other runs' files in `/tmp` and leftover work directories, or called the `/rank` endpoint directly. These runs held 73–99% of each condition's tokens, so condition means mostly measured how much an agent explored the machine.

We fixed the first by plugging the DM into Hermes's own bridge (same compact hits, `tool_describe` and typed calls; only the ranking differs) and the second by per-run mount-namespace isolation and per-run MCP tokens (§4).

Isolated, paired comparison. In the isolated run (`hermes_v3`: 12 v1 tasks, two repeats, DeepSeek V4.1 Flash, 613 tools) all three conditions passed 23 of 24 runs. Paired by task and repeat (Table 5, Figure 3), the DM in Hermes's bridge used 0.88× the tokens of Hermes's BM25 (geometric mean of per-slot ratios; 95% bootstrap CI over tasks 0.77–1.02), cost $0.0107 against $0.0126 per run, searched 2.8 times per run against 4.2, and used fewer tokens on 17 of 24 slots. DM v2 (exact-name shortcut and 90% mass cutoff) was not better: 0.93× [0.79, 1.11]. A random ranker in the same bridge, measured in an earlier run before isolation, used about 1.8–2× the tokens of BM25: ranking matters, but between two reasonable rankers the difference is small. Restricted to runs that stayed on task, the rankers tie (for example, median 215k against 240k tokens).

Noise. Rerunning the same task in the same configuration changed its token count by a median factor of 1.45 (standard deviation of the log ratio 0.37), and the paired log ratio between two conditions had a standard deviation of 0.44. At that variance about 136 pairs are needed to detect a 10% difference in tokens with 80% power at $\alpha = 0.05$, several times what this run had. Three sources contribute: OpenRouter routed DeepSeek requests to 48 endpoints with different quantizations (fp4, fp8, mxfp4); the harness sets no fixed temperature; and agents wander, re-reading files and re-running searches in different orders.

Table 5: Hermes Agent, isolated run (`hermes_v3`): 12 v1 tasks × 2 repeats, DeepSeek V4.1 Flash, 613 tools. Ratio: geometric mean of per-slot token ratios against Hermes BM25, with a 95% bootstrap CI that resamples tasks.

Ranker in Hermes's bridge	Pass	Tokens (median)	$/run	Searches	Ratio [95% CI]	Fewer tokens
BM25 (Hermes default)	23/24	325k	0.0126	4.2	1	–
DM	23/24	299k	0.0107	2.8	0.88 [0.77, 1.02]	17/24
DM v2	23/24	327k	0.0128	5.1	0.93 [0.79, 1.11]	13/24

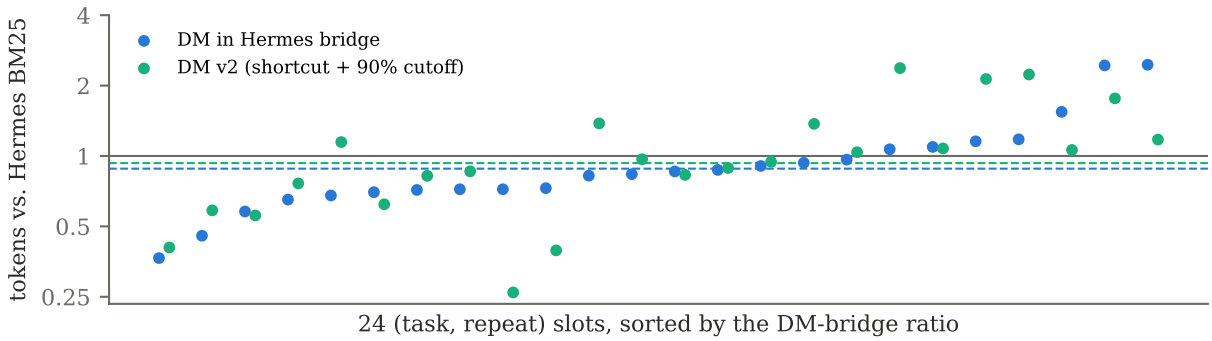


Figure 3: Per-slot token ratio against Hermes's BM25 in the isolated Hermes run (log scale). Dashed lines: geometric means (0.88 and 0.93).

5.4 Final controlled run

The final experiment is designed to remove the confounds above and to come close to the power needed to detect a 10% token difference (114 paired slots per comparison, against the roughly 136 estimated above). Its settings:

- Hermes Agent with per-run isolation and its `approvals.deny` list blocking probing commands (`curl`, `wget`, `ps`, `ss`, `netstat`, `/proc`, `localhost`, ...);
- task set v2 (38 tasks) on the 1,076-tool catalog, with schema validation of arguments;
- one LLM, DeepSeek V4.1 Flash, pinned to DeepInfra's fp8 endpoint, temperature 0.6, seed 7;
- every prompt prefixed with the same note on connected services (§4);
- four conditions: Hermes BM25, the DM in Hermes's bridge, a random ranker in the same bridge, and native loading of all definitions;
- three repeats: $38 \times 4 \times 3 = 456$ runs, queued repeat by repeat, with all four conditions of a (task, repeat) slot queued back to back so that paired runs see the same provider conditions;
- paired analysis with task-level bootstrap CIs; pass@1 and pass^3 (all three repeats pass).

We also piloted a second LLM, Kimi K3.1 Pro, and dropped it for cost and speed: one unfinished smoke run cost about $0.44, and the pinned endpoints generated about 2 (DeepInfra bf16) and 22 (Moonshot first-party) tokens per second. Table 6 and Figure 4 will report it.

Results. [TODO: fill in from the final run: pass@1 and pass^3 per condition; DM vs. BM25 token ratio with CI and number of pairs; whether the random control and native separate from BM25; failures by type (wrong server including decoys, destructive calls, unfinished multi-service chains);

Table 6: Final controlled run: Hermes Agent, DeepSeek V4.1 Flash (pinned), task set v2 (38 tasks), 1,076 tools, three repeats (114 slots per condition). Tokens: median per run; ratio: geometric mean of per-slot token ratios against BM25 with a task-level bootstrap CI; fewer: slots on which the condition used fewer tokens than BM25. **[pending: all cells]**

Condition	pass@1	pass^3	Tokens	$/run	Searches	Ratio [95% CI]	Fewer
Hermes BM25	**[pending]**	**[pending]**	**[pending]**	**[pending]**	**[pending]**	1	–
DM in bridge	**[pending]**	**[pending]**	**[pending]**	**[pending]**	**[pending]**	**[pending]**	**[pending]**
random in bridge	**[pending]**	**[pending]**	**[pending]**	**[pending]**	**[pending]**	**[pending]**	**[pending]**
native	**[pending]**	**[pending]**	**[pending]**	**[pending]**	–	**[pending]**	**[pending]**

[pending: figure]
Per-slot token ratio DM vs. BM25 and random vs. BM25, and pass rate by task category (single, multi, absent, control) for each condition.

Figure 4: Final controlled run. **[pending]**

behaviour on absent tasks; how often the deny list fired. Update the abstract, introduction and conclusion accordingly.]

6 Threats to validity and lessons

Ceiling. On v1 every search mechanism passed nearly every run, so pass rate cannot rank them; we rank by tokens and cost, which are noisy. The v2 tasks (symptom-first prompts, more multi-service tasks, decoy servers) are meant to lower the ceiling; whether they do is **[pending]**.

Simulated answers. Definitions are real, answers are not. Real services return longer, messier results, which would raise every condition's token count and could change how agents search. Unmatched reads return empty results, which is realistic but may make absent tasks easier to recognize than with a real service.

The connected-services note. Every prompt of the final run starts with the same note on connected services (§4), in all conditions including native. Real harnesses put a comparable statement in the system prompt, and our synthetic pilot showed that agents not told what exists may never search (App. B). The note names categories of services, not brands, so it does not tell the agent which server to search; it may still raise search rates above what an unprompted agent would show.

Approximated mechanisms. We reimplemented Claude Code's ToolSearch, Copilot's virtual tools and Goose's routers from their documentation rather than running those harnesses; only Hermes's bridge is the shipped code. Our cc scoring in particular is an approximation.

Query distribution and scale. Hand-written retrieval queries overstate the DM's advantage; agent-written queries are shorter and more lexical (§5.1). The live results so far rest on two inexpensive LLMs, one DM and, for Hermes, 24 paired slots; the final run uses a single LLM.

Lessons for measuring agents. (1) *Sandbox the shell, not only the tools.* Harness-level permissions did not stop agents from reading what they should not; in Hermes, agents probing the machine dominated the token counts. Per-run isolation of files, `/tmp`, home directory and service tokens was necessary. (2) *Separate packaging from ranking.* Plugging rankers into the same bridge changed the conclusion from "the DM costs more" to "the DM may cost slightly less". (3) *Pin providers and sampling.* Routing across dozens of endpoints with different quantizations adds variance that is expensive to average away. (4) *Pair runs.* With a paired log-ratio standard deviation of 0.44, even paired designs need over a hundred pairs to detect a 10% token difference; unpaired comparisons need more.

7 Related work

Tool retrieval for large catalogs. Retrieving tools before generation is the standard answer to large catalogs [7, 23]. RAG-MCP [10] retrieves MCP tool descriptions with embeddings to reduce prompt bloat, and MCP-Zero [9] lets the agent request tools actively by describing what it needs, server first and tool second, the same hierarchy our DM uses. Lee [12] find that selecting a tool is largely retrieval. We compare these ideas with the mechanisms harnesses actually ship, and with a DM, inside unmodified agents.

MCP benchmarks. LiveMCPBench [16] evaluates agents that must navigate a large set of real MCP servers, and MCPMark [27] stress-tests realistic MCP use with verifiable tasks; MCP-Atlas [5] and Toolathlon [13] also score task completion with real tools. They measure agents' end-to-end success; we hold the agent fixed and vary only the selection mechanism, with simulated back ends so that runs are reproducible and writes can be graded safely.

[TODO: verify: vendor-doc URLs in the bibliography (Claude Code, Anthropic tool search, Agent Skills, Copilot virtual tools, Codex CLI, goose) and the author lists of LiveMCPBench and MCPMark were written from memory.]

Vendor reports. Stacklok compared its MCP Optimizer with Anthropic's tool search tool on 2,792 tools and reported much higher selection accuracy for its hybrid retrieval [25]; Arcade tested Anthropic's tool search on about 4,000 tools [11]. Both measure selection accuracy rather than agents completing tasks, and neither reports token costs per task.

Decision models. The Artificial Analysis Decision Index [4] and an analysis of it [19] compare DMs with LLMs on standalone decisions, including tool routing. We place a DM inside agents, as one component of tool search.

8 Conclusion

A decision model can serve as a coding agent's tool selector. On real MCP tool definitions it ranks tools more accurately than the lexical and embedding searches that current harnesses ship, and more cheaply than an LLM router, and a server-then-tool hierarchy takes it past its 255-option limit. Whether this makes agents better is a different question. In live runs every search mechanism, the DM included, reaches ceiling pass rates and removes most of the context that loading all tools costs; within Hermes's own bridge the DM used about 12% fewer tokens than BM25, a difference our pilot cannot yet distinguish from none. The packaging of search results mattered more than

their ranking. The final controlled run ([**pending**]) will show whether the DM's ranking advantage survives harder tasks, a larger catalog with decoys, and a pinned provider.

Reproducibility. The tool harvester, the catalog snapshots, the simulator, the 71 tasks with hidden tests and reference solutions, the MCP server with all selection conditions, the Hermes patch, the sandbox setup, all run records and the analysis and figure scripts will be released.

Use of AI assistance. Claude Code, Anthropic's coding agent, helped throughout this work: it wrote the harness, catalog harvester, tasks and analysis scripts, ran the experiments, and drafted the paper, under the authors' direction. The authors chose the research questions, reviewed the code, results and text, and are responsible for the content. The session analysed in §3 is this assistant's own session. [TODO: authors: confirm or revise the AI-use statement and the author list.]

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A Sandbox details

opencode. The agent runs as an unprivileged user with an empty environment in a fresh copy of the task. Permissions deny everything except reading and editing inside the project, the to-do list, the catalog tools, and shell commands matching `pytest`. Experiment files are readable only by root; the API key is held by a local proxy that admits only the experiment models. We scanned every transcript for file access outside the task folder. A first pilot that relied on harness permissions alone was discarded: the agent read hidden tests and answer files through absolute paths.

Hermes. Hermes runs one-shot (`hermes -z "<prompt>" -yolo`); with interactive approvals off, the sandbox is the guard. From `hermes_v3` on, each run executes in its own mount namespace with a private work directory, home directory and `/tmp`; the MCP server accepts only requests that carry the run's token, which Hermes reads from its environment and strips from its shell's. The final run adds Hermes's `approvals.deny` list for probing commands, as an organization policy would; the isolation remains the actual guard.

B The earlier synthetic pilot

An earlier version of this study used a fictional company's catalog of 100 and 300 synthetic tools (about 610 tokens each) with opencode, and compared loading all tools, a DM-backed `find_tool`/`call_tool` pair with and without a service list, and tools preselected from the task prompt. It showed the same pattern as the real-catalog runs: search matched native loading at a fraction of the context, and an agent without a list of the services sometimes never searched. We replaced it because its templated descriptions were more uniform and easier to rank than real ones, its tasks were at ceiling, and its tools resembled no real server, so it could not compare the mechanisms that harnesses ship. Its code and results remain in the repository.

C Task set v2

Table 7: Task set v2 (38 tasks). Prompts are shortened to their first sentence and shown without the escape clause. Services: the servers of the needed tools, in order; $+w$: number of required writes; $\star$: new in v2.

Category	Task	Prompt (first sentence)	Services
control	`c_chunk`	chunk in batching/chunk.py drops the last partial chunk.	–
control	`c_dedupe`	dedupe in collections_util/dedupe.py loses the original order.	–
control	`c_duration`	parse_duration in timeutil/duration.py only understands hours.	–
control	`c_leap`	is_leap in calendar_util/leap.py is wrong for some years.	–
single	`confluence_limits`	Our public API rate limits changed.	atlassian
single	`eks_importer` $\star$	The data team's nightly batch-importer job has failed three nights in a row.	aws_eks
single	`gh_ci_fail`	Pull request #214 in acme/payments can't be merged because a check is red.	github $+1w$
single	`gh_codesearch`	SUPPORTED_CURRENCIES in shop/currencies.py has drifted from the company-wide canonical list, which lives in another repository.	github
single	`gh_dependabot`	Security says acme/payments has dependencies with known vulnerabilities.	github
single	`gh_issue`	Please implement acme/payments#301.	github
single	`gh_open_pr`	Pagination in acme/catalog is broken (see #57).	github $+2w$
single	`gh_release`	We are behind on the ledger API.	github
single	`gh_review`	There is reviewer feedback waiting on pull request #88 in acme/ledger.	github $+1w$
single	`grafana_incident`	We are in the middle of an incident about checkout timeouts and the responders agreed on a mitigation.	grafana $+1w$
single	`jira_impl`	Pick up PAY-412: implement it, then move it to review.	atlassian $+1w$
single	`k8s_crashloop`	ledger-worker keeps restarting in production (namespace ledger) since the last config change.	kubernetes
single	`k8s_oom`	report-builder (namespace reports) keeps dying in production.	kubernetes
single	`linear_bug`	Fix ENG-77 and move it to In Review.	linear $+1w$
single	`loki_errors`	Customers stopped getting shipping e-mails at around 06:00 UTC today; the notifier service is the suspect.	grafana
single	`mongo_shape`	The nightly export has started crashing on recent orders.	mongodb
single	`notion_runbook`	We are rotating the webhook signing keys.	notion
single	`oncall_contact`	Set ESCALATION_CONTACT in ops/escalation.py to the e-mail address of whoever is on call for the payments team right now.	grafana

Category	Task	Prompt (first sentence)	Services
single	`pg_migration`	Add migrations/0042_payments_merchant_idx.sql with an index named payments_merchant_created_idx for looking up a merchant's payments by creation time.	postgres
single	`prom_threshold`	Set SLOW_REQUEST_MS in alerts/config.py to twice the checkout service's current p99 request latency, in milliseconds.	grafana
single	`sentry_events`	Since this morning's deploy some clients get HTTP 500 from POST /refunds.	sentry
single	`sentry_fix`	The noisiest production error of payments-api right now needs to go.	sentry +1w
single	`slack_decision`	The payments team agreed on a new webhook retry policy in a chat discussion.	slack +1w
multi	`cw_reports` ⋆	The legacy reports API has been answering 502 since yesterday evening.	aws_cloudwatch, slack +1w
multi	`gl_pipeline` ⋆	The main-branch pipeline of billing-legacy is red.	gitlab +1w
multi	`linear_notion_pr` ⋆	Do ENG-91 the way our API guidelines say, and open a pull request for it from a branch named eng-91 against main in acme/public-api.	linear, notion, github +1w
multi	`multi_alert`	Something is paging for the checkout service.	grafana, slack +1w
multi	`multi_jira_conf`	Implement PAY-530 as its design document specifies, then leave a one-line summary on the ticket.	atlassian +1w
multi	`multi_sentry_gh`	A recent change to ledger-worker introduced a reconciliation bug that shows up as a production error.	sentry, github +1w
multi	`promo_sentry_jira` ⋆	Android users report they can't apply promo codes.	sentry, atlassian +1w
absent	`absent_jira`	Implement PAY-999 in payments/limits.py.	–
absent	`absent_release`	Pin SDK_VERSION in sdk/version.py to the newest release of acme/ghost-sdk (without the leading v).	–
absent	`absent_sentry`	The billing-cron error someone mentioned in standup needs fixing in billing/cron.py.	–
absent	`absent_slack`	Set CACHE_TTL_S in cache/config.py to the value the platform team agreed on.	–